

Pulse

<https://d1kvjmmokxigdv.cloudfront.net>

A pulse reading from a phone camera that refuses to answer when it cannot answer well.

Most smartphone pulse recordings are not measurable

A fingertip over a lit camera brightens and dims with each heartbeat, and reading a rate off that is straightforward. Knowing when the reading is wrong is the whole problem. In BUT PPG — 3,888 smartphone recordings, each with a simultaneous ECG — expert annotators marked only 795 usable. A tool that always returns a number is therefore wrong most of the time, confidently.

What it does

Pulse resamples the camera signal to a fixed 30 Hz, filters it, finds beats, and slides an eight-second window to measure the calmest stretch. Then four checks decide whether to answer at all. The one that does the real work requires beat-counting and the frequency spectrum — two independent estimates — to agree within 5%, which is what catches the half-rate errors a moving finger produces.

What it measures

Recordings analysed	3,840
Refuses what expert annotators refuse	98.1%
Recordings accepted	202 (5.3%)
Rate within 5 bpm of ECG	85.1%
Median absolute error	0.93 bpm
Mean absolute error	6.11 bpm

Scored against heart rates from the ECG's own R-peak annotations, not the dataset's HR column, which is doubled for one subject. Accepting 5.3% is the price of not reporting a wrong rate. The mean exceeds the median because a few accepted recordings are badly wrong rather than slightly wrong — that tail is the honest weakness and it is reported.

What it does not do

- Does not diagnose anything. It reports a rate, or it reports nothing.
- Not an atrial fibrillation screen: the corpus has no rhythm labels, so the irregularity measures are described and never interpreted.
- Validated only on ten-second recordings. The app records thirty; that the accuracy trend continues is an extrapolation, not a measurement.
- Nothing characterised across skin tones or camera hardware.

Sources. BUT PPG 2.0.0, PhysioNet, CC-BY · browser and Python verified identical across 418 traces and 1,570 windows (max divergence 0.0 on every scalar)

Full citations, including figures rejected as unverifiable, in docs/pulse/sources.md